

Text Files: Reading

To read a file, open it with `r` mode. In general, it is not a good idea to use **fscanf** for reading: Unless the file is perfectly formatted, `fscanf` will not handle it correctly. Instead, use **fgets** to read in each line and then parse out the pieces you need.

The following code demonstrates the process of reading a file and dumping its contents to the screen:

```
#include <stdio.h>

int main()
{
    FILE *f;
    char s[1000];

    f=fopen("infile","r");
    if (!f)
        return 1;
    while (fgets(s,1000,f)!=NULL)
        printf("%s",s);
    fclose(f);
    return 0;
}
```

The `fgets` statement returns a `NULL` value at the end-of-file marker. It reads a line (up to 1,000 characters in this case) and then prints it to `stdout`. Notice that the `printf` statement does not include `\n` in the format string, because `fgets` adds `\n` to the end of each line it reads. Thus, you can tell if a line is not complete in the event that it overflows the maximum line length specified in the second parameter to `fgets`.

C Errors to Avoid

- Do not accidentally type **close** instead of **fclose**. The `close` function exists, so the compiler accepts it. It will even appear to work if the program only opens or closes a few files. However, if the program opens and closes a file in a loop, it will eventually run out of available file handles and/or memory space and crash, because **close** is not closing the files correctly.